

JL_2^* AND PROBLEM 1 OF BARROSO–KALENDA–LIN

RUN FA_BANACH_001

Status. `literature_implied_answer` (negative answer to Problem 1; Problem 2 not claimed). This note records an existing literature implication, not a new theorem of the run.

SOURCE QUESTION

In *On the approximate fixed point property in abstract spaces*, arXiv:1101.5274, Barroso–Kalenda–Lin ask the following as Problem 1: for an arbitrary Hausdorff locally convex space X , is it true that every bounded sequence in X has a weakly Cauchy subsequence if and only if each bounded separable subset of X is Frechet–Urysohn in the weak topology? They point out that the answer is positive for locally convex spaces admitting a compatible metrizable locally convex topology, and they suggest (X^*, w^*) for a Johnson–Lindenstrauss space as a possible source of a negative answer.

SUPPORTING LITERATURE

Martinez-Cervantes, *Banach spaces with weak*-sequential dual ball*, arXiv:1612.05948, proves that the classical Johnson–Lindenstrauss space JL_2 has weak-star sequential dual ball of sequential order 2 (Theorem 3.1 in that paper). The paper also recalls that weak-star sequential dual ball implies weak-star sequential compactness of the dual ball, and that JL_2 does not have weak-star angelic dual. Equivalently, $(B_{JL_2^*}, w^*)$ is sequentially compact and sequential, but it is not Frechet–Urysohn.

IDENTIFICATION

Let $Z = JL_2$, and set

$$E = (Z^*, \sigma(Z^*, Z)).$$

This is a Hausdorff locally convex space. Its continuous dual is Z , so the weak topology of E is again $\sigma(Z^*, Z)$.

First, every bounded sequence in E has a weakly Cauchy subsequence. Indeed, if (x_n^*) is bounded in E , then it is pointwise bounded on Z . By the uniform boundedness principle, (x_n^*) is norm bounded in the Banach dual Z^* . After rescaling, the sequence lies in B_{Z^*} . Since (B_{Z^*}, w^*) is weak-star sequentially compact, some subsequence converges in $\sigma(Z^*, Z)$. This

subsequence is weakly convergent, hence weakly Cauchy, in the locally convex space E .

Second, not every bounded separable subset of E is Frechet–Urysohn in the weak topology. Since (B_{Z^*}, w^*) is not Frechet–Urysohn, there are $A \subset B_{Z^*}$ and $p \in \overline{A}^{w^*}$ such that no sequence in A converges to p . Since the dual ball is sequential, it has countable tightness. Therefore there is a countable $A_0 \subset A$ with $p \in \overline{A_0}^{w^*}$. Let $S = A_0 \cup \{p\}$. The set S is countable, hence separable, and is bounded in E . In the subspace S , the point p lies in the closure of A_0 , but no sequence from A_0 converges to p . Thus S is not Frechet–Urysohn.

Consequently the two properties in Problem 1 are not equivalent for arbitrary Hausdorff locally convex spaces.

SCOPE

This does not settle Problem 2 of Barroso–Kalenda–Lin, which asks whether weak-AFPP is equivalent to the bounded-sequence weakly-Cauchy-subsequence property for arbitrary Hausdorff locally convex spaces.

This packet is placed under literature-implied answers because arXiv:1612.05948 does not explicitly state that it answers arXiv:1101.5274, but its theorem does so after the direct locally convex identification above.

REFERENCES

REFERENCES

- [1] C. S. Barroso, O. F. K. Kalenda, and P.-K. Lin, *On the approximate fixed point property in abstract spaces*, arXiv:1101.5274.
- [2] G. Martínez-Cervantes, *Banach spaces with weak*-sequential dual ball*, arXiv:1612.05948.